

Machine Learning Approaches to Human Activity Recognition: A Literature Review

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I. INTRODUCTION

Human Activity Recognition (HAR) is the task of automatically identifying what physical activity a person is doing, such as walking, sitting, running, or climbing stairs using data from wearable sensors. These sensors, called Inertial Measurement Units (IMUs), include accelerometers (which measure movement speed) and gyroscopes (which measure rotation). Modern smartphones and fitness trackers contain these sensors, making HAR technology widely accessible for applications in healthcare monitoring, elderly care, sports training, and rehabilitation [1].

HAR is a supervised classification problem in machine learning. This means we train the computer to recognize activities by showing many examples of sensor data where we already know which activity was performed. The sensor data is divided into short time windows (usually 1 to 5 seconds), and each window gets labeled with one activity. The challenge is teaching the computer to tell the difference between similar activities, for example, walking versus jogging; at the same time working perfectly for new users.

Researchers use standard datasets to test and compare different methods. The most popular is the UCI Human Activity Recognition (UCI-HAR) dataset, which contains data from 30 people performing six activities while wearing a smartphone [3]. Other datasets include WISDM (36 people, smartphone), PAMAP2 (9 people wearing sensors on wrist, chest, and ankle), and HARTH (22 people performing 12 everyday activities in natural conditions) [2]. These datasets range from simple laboratory tests to complex real world situations, presenting different levels of difficulty for machine learning models.

This review examines seven recent research papers (2021-2025) that use machine learning to solve the HAR problem. We focus on comparing traditional machine learning methods such as Support Vector Machines (SVM), Random Forest, k-Nearest Neighbors (k-NN), and XGBoost with newer deep learning approaches including Convolutional Neural Networks (CNN) and Long Short-Term Memory networks (LSTM). Our goal is to understand which methods work best, why they work, and what trade-offs exist between different approaches.

II. METHODS AND ANALYSIS

A. The HAR Processing Pipeline

All the studies we reviewed follow similar steps to process the sensor data and make predictions. First, raw sensor signals are cleaned using filters to remove noise and separate body movement from gravity. Then, the signals are divided into overlapping time windows. The key difference between methods comes next: traditional machine learning requires humans to manually calculate statistical features from each window (such as average, standard deviation, maximum value, and correlation between sensor axes), while deep learning methods can learn patterns directly from the raw sensor data without this manual feature calculation step.

B. Traditional Machine Learning Approaches

Logacjov et al. [2] tested four traditional machine learning methods on their HARTH dataset using 161 hand calculated features: k-Nearest Neighbors (k-NN), Support Vector Machine (SVM), Random Forest, and XGBoost, each method explained subsequently:

k-Nearest Neighbors (k-NN) is one of the simplest machine learning algorithms. To classify a new activity window, k-NN looks at the k training examples that are most similar (measured by Euclidean distance in feature space) and assigns the majority class. For example, if $k=5$ and three of the five nearest neighbors are 'walking' while two are 'running', the new window is classified as 'walking'. The value of k controls the decision: small k values (like $k=3$) make the model sensitive to noise but capture detailed patterns, while large k values (like $k=15$) create smoother decisions but may miss subtle differences. The optimal k is found through cross validation. While k-NN requires no training phase, it must compare new data against every training example during prediction, making it computationally expensive for large datasets.

Support Vector Machine (SVM) finds the best boundary (called a hyperplane) that separates different activity classes with the maximum margin. When activities cannot be separated by a straight line, SVM uses a kernel function typically the Radial Basis Function (RBF) to map the data into a higher dimensional space where separation becomes possible. This is like viewing scattered 2D data from a 3D angle where patterns become clear. In the HARTH study, SVM achieved the best performance with an F1 score of 0.81 when tested on completely new people (leave one subject out evaluation), outperforming all deep learning models tested [2].

Random Forest is an ensemble method that combines many decision trees, where each tree is trained on a random subset of features and data. This is similar to asking multiple experts who each see slightly different information, then taking a majority vote on their predictions. Balaha and Hassan [3] showed that Random Forest achieved 100% accuracy on UCI-HAR and 99.10% on WISDM dataset. The ensemble approach makes Random Forest robust and resistant to overfitting compared to single decision trees.

XGBoost (Extreme Gradient Boosting) is another ensemble method, but unlike Random Forest which builds trees independently, XGBoost builds trees sequentially. Each new tree tries to correct the mistakes made by previous trees, gradually improving predictions through this boosting process. XGBoost which is an advanced ensemble technique alongside Random Forest (bagging) and AdaBoost (boosting).

Importantly, Balaha and Hassan [3] also tested dimensionality reduction before classification. They used Principal Component Analysis (PCA) to reduce the 561 features in UCI-HAR to fewer dimensions while retaining the most important information. This PCA preprocessing step improved efficiency without sacrificing accuracy when paired with SVM and Random Forest classifiers.

C. CNN LSTM Hybrid Architectures

Deep learning models learn features automatically from raw data rather than requiring manually calculated statistics. The two fundamental architectures for HAR are Convolutional Neural Networks (CNN) and Long Short-Term Memory Networks (LSTM).

CNNs apply sliding filters across the sensor data to detect short patterns in movement, similar to how your eyes recognize edges and shapes in images. For HAR, 1D CNN filters slide across the time dimension to identify bursts of acceleration that characterize specific activities. For example, a filter might learn to detect the sharp acceleration spike that occurs during each running step.

LSTMs are designed to remember patterns across time. They use gating mechanisms to input gates (what new information to remember), forget gates (what old information to discard), and output gates (what to pass forward) to capture how activities evolve over seconds. This makes LSTMs effective for HAR because activities have temporal structure: walking consists of repeated step cycles, and sitting is characterized by sustained stillness.

Several studies combined CNN and LSTM into hybrid architectures to leverage both strengths. Khatun et al. [4] built a model where CNN first extracts local motion patterns, then LSTM processes these patterns across time to recognize complete activities. This architecture achieved 98.76% accuracy on the MHEALTH dataset. However, on UCI-HAR it only reached 93.11%, lower than the 100% achieved by traditional Random Forest [3], showing that deep learning does not always outperform simpler methods.

Luwe et al. [5] used a variant called Bidirectional LSTM (BiLSTM), which reads the time window both forwards and backwards. This captures future context as well as past information, which helps window boundaries where activity transitions occur. Their model achieved 95.48% accuracy on UCI-HAR. While BiLSTM and attention mechanisms represent advances, we include them here to show how researchers build the fundamental CNN and LSTM architectures taught in the module.

III. DISCUSSION AND EVALUATION

A. Performance Summary

Table I summarize the best results from each study. The performance numbers vary significantly depending on which dataset was used and how the model was tested.

TABLE I. Performance Comparison Across Reviewed Studies

Study	Method	Dataset	Best Result
Logacjov [2]	SVM + Features	HARTH (real world)	F1 = 0.81
Balaha [3]	Random Forest	UCI-HAR (lab)	100% Acc

Khatun [4]	CNN LSTM	MHEALTH	98.76% Acc
Luwe [5]	CNN BiLSTM	UCI-HAR	95.48% Acc

B. Traditional ML Versus Deep Learning

The results show that neither approach is universally superior; it depends on the situation. On the UCI-HAR dataset, both Random Forest and SVM achieved near perfect accuracy (100% and 99.10% respectively) [3]. This demonstrates that when features are carefully designed and the problem is well defined, traditional machine learning can match or exceed deep learning performance.

However, the HARTH dataset tells a different story. This dataset contains 12 activities recorded in natural daily life rather than in a laboratory. When tested on completely new people (leave one subject out testing), SVM achieved only 0.81 F1 score [2], meaning it correctly classified about 81% of activities. The high variability (± 0.18) reveals that the model performs very differently for different people; some participants are recognized well, while others are not.

The key lesson: traditional ML methods (SVM, Random Forest, k-NN, XGBoost) work very well on clean, controlled datasets where good features can be engineered. Deep learning methods (CNN, LSTM) become more valuable when: (1) there is lots of training data available, (2) the activities are complex with subtle patterns, and (3) manually designing features would be difficult.

C. The Evaluation Protocol Problem

An important finding from this review is that how you test a model dramatically affects the reported accuracy. Most studies [3][4][5] used random train test splits, where data from the same person appears in both training and testing sets. This can inflate accuracy by 5-15% compared to leaving one subject out (LOSO) testing, where the model is tested on new users during training [1][2]. Random splits make results look better but do not reflect real world performance where the model must work on new users. This inconsistency makes it difficult to compare studies and represents a major weakness in current research. Future work should standardize LOSO testing to get honest performance estimates.

D. Architectural Trade-offs and Deployment Considerations

While deep learning models generally achieve higher accuracy on complex datasets, they come with significant computational costs. SVM and Random Forest models can make predictions in microseconds on simple processors, making them suitable for smartphones and fitness trackers. In contrast, CNN LSTM and more advanced models require much more computation time, memory, and battery power.

Most of the reviewed studies do not report energy consumption or inference speed to a critical oversight since wearable HAR applications run on battery powered devices with limited processing capability. This trade-off between

accuracy and computational efficiency represents the main practical challenge when choosing between traditional ML and deep learning for real world deployment.

IV. CONCLUSION

This review examined seven recent research papers comparing machine learning approaches for Human Activity Recognition. The main finding is that traditional machine learning methods (SVM, Random Forest, k-NN, XGBoost) can achieve excellent performance when the problem is well defined, and features are carefully designed. On the UCI-HAR benchmark, Random Forest achieved 100% accuracy [3], matching the performance of much more complex deep learning models.

Deep learning methods, particularly CNN and LSTM, become advantageous when dealing with complex, real world data where manually designing features is difficult. Hybrid CNN LSTM models [4][5] can automatically learn both local motion patterns (via CNN) and temporal sequences (via LSTM) without human feature engineering. However, this comes at the cost of requiring more training data, longer computation time, and higher energy consumption.

Three major limitations emerged from the review. First, inconsistent evaluation methods (random splits versus leave one subject out testing) make it difficult to compare studies fairly; random splits can inflate accuracy by up to 15% [2]. Second, computational cost and energy consumption are rarely reported despite being critical for wearable deployment [6][7]. Third, how well models work for new users remains poorly quantified, with the HARTH study showing that even the best SVM model still misclassifies one in five activities when tested on unseen participants [2].

Based on these findings, we recommend three specific research directions. First, the HAR community should adopt leave one subject out testing as standard practice, as demonstrated by Logacjov et al. [2], to provide realistic performance estimates for new users. Second, studies should report inference time and energy consumption alongside accuracy, particularly when proposing deep learning models like the CNN LSTM hybrids in Khatun et al. [4] and Luwe et al. [5], to enable informed deployment decisions. Third, personalization techniques that adapt models to individual movement patterns should be investigated to address the high variability (± 0.18 F1 score) observed in LOSO evaluation [2], moving beyond the one size fits all approach dominant in current research.

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